TTS v2: Speech-model-v4-TTS

Male-voice first word · youthful voices · 7-emotion control · lower latency. ThonburianTTS (F5-TTS 337M) → v3 → v4. All numbers in this report are computed from files in TTS-Lab/data/; automatic metrics only (no MOS/CMOS: no human has rated anything yet).

1. Summary and acceptance

criterion	verdict	evidence
1 Male first word	PARTIAL	Reported sentence (3 seeds): v3 as shipped male 0.00/0.33 (turbo/Pathumma) → v4 male 0.33/1.00. Broad set (Pathumma): male 0.987, female 1.000; turbo male 0.855 vs female 0.868 (male not lower). Clipped onsets 0. The gain comes mostly from the new reference voices and the 0.3 s tail, not from training.
2 CER / similarity / gap	MET	paired CER v4-v3 -0.0096 (95 % CI -0.036..+0.016); speaker similarity -0.0019 (-0.011..+0.007); longest internal pause of gap sentences 0.63 s (v3 0.41 s).
3 Youthful (18-23)	NOT MET	estimated age of the generated default voices: median 26.0 (male) / 26.6 (female) years; share in 18-23: 18% / 8%. The four references are estimated at 23.6-24.3. Listening not done.
4 Emotion	NOT MET by the model alone	independent SER accuracy 0.152 at intensity 1.0 (chance 0.143); F0 moves < 0.5 semitone (templates 1-3.5); intensity monotonic for 2 of 6 emotions. With the PSOLA layer: SER 0.20 / 0.25 (intensity 0.5 / 1.0) and the prosody follows the templates by construction. CER not degraded (0.006-0.032).
5 Latency	MET (cold start barely)	Docker + gateway, RTX 2050, p50: 1 s 3.21 → 1.25 s, 3 s 4.80 → 2.12 s, 6 s 6.55 → 2.84 s (2.3-2.6x). Cold start 72 → 68 s. Plus ~5 s per request removed in the sign-api (dead TTS host) and text returned before audio.
6 Cost	MET	GPU from job records $3.11; project estimate ≈ $3.6 incl. cross-region reads and storage (not visible in the job records). Ceiling $4.

What did NOT improve: the model's own emotion control (pitch barely moves after a 20-minute run), the perceived age target, and a measurable first-word gain from training itself. What did: the reported first-word failure (reference choice + silence tail + prompt-format training), speed (2.3-2.6x at equal intelligibility), intelligibility under emotion, and a production latency bug.

2. Baselines (v1/v3)

v3 = ThonburianTTS fine-tuned in v1 (12.4 h S+, 2,774 updates; report TTS-result-v1.pdf). Baselines are re-measured here: 'v3 as shipped' (old references, 50 ms tail, 32 steps; local runs in data/ablation) and 'v3 weights with the new tail/voices' (cloud run 1, A100). The cloud v3 rows already use the new 0.30 s tail, so they understate v3 as shipped; both are reported.

3. Data: what was used and how

Source: Chalermdej/yodas2_sidon_th_tts, all three grades. Metadata (141,927 clips, 156 h) was filtered before any audio was downloaded: duration 2-12 s, text agreeing (difflib ratio ≥ 0.9) with BOTH whisper_large and whisper_typhoon, DNSMOS per grade (S+ ≥ 3.2, S ≥ 3.3, A+ ≥ 3.3; DNSMOS barely differs between grades: median 3.33-3.34 in all three, so the grade is not a noise label), duplicate text removed, ≤ 40 clips per speaker id. That selected 61,509 clips (73.4 h, 3,891 anonymised speakers); all 56 shards (26.6 GB) were downloaded, decoded and deleted shard by shard; no clip failed the clipping / silence checks.

Annotation passes (local RTX 2050): perceived age (audeering/wav2vec2-large-robust-24-ft-age-gender; at most 6 clips per speaker, speaker median) and emotion posteriors (emotion2vec_plus_large) on exactly 30,000 randomly ordered clips (the pass was stopped at the number you asked for, 30,000). The training pool = those labelled clips minus test speakers = 29,162 clips, 34.7 h, 3,470 speakers, 3.2 GB of FLAC.

grade	voice presentation	clips	hours
A+	feminine	316	0.4
A+	masculine	358	0.4
A+	unknown	86	0.1
S	feminine	3149	4.4
S	masculine	3499	4.5
S	unknown	666	0.9
S+	feminine	10921	12.7
S+	masculine	9164	10.3
S+	unknown	1841	2.0

Voice presentation is a preliminary acoustic label: speaker-median F0 (≥ 180 Hz feminine, ≤ 150 Hz masculine) that must agree with the age model's gender head; 150-180 Hz is decided by the head only when it is > 0.9; otherwise 'unknown'. A bug worth reporting: the gender head's output order is (female, male, child), not (child, female, male); it was found because male-F0 speakers scored 0.97 'female' and was verified against F0 before any label was used.

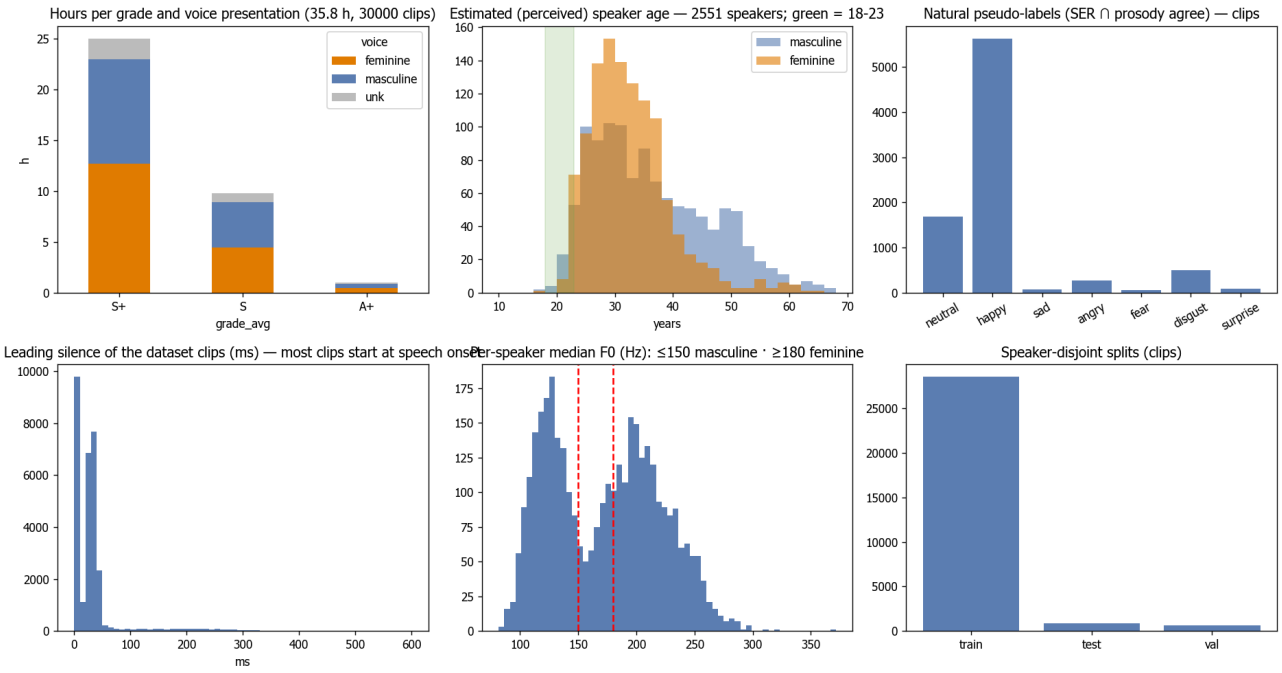


Figure 1. Data composition: hours per grade and presentation, estimated speaker age, natural emotion pseudo-labels, leading silence, F0, splits.

Speaker leakage: splits are speaker-disjoint (test 838 clips, val 645); v1's test and val speakers stay in their splits so v3 and v4 are compared on the same held-out speakers; the four default-voice reference speakers were moved to test. Speaker ids are YouTube video ids and are not verified identities (one video can hold several people, one person several videos); the manifest only stores a salted hash, and no raw id is published.

Leading silence: the dataset clips are cut tightly at speech onset (median leading silence 20 ms, 25 % of clips 0 ms), which is why the model learns to start speech at the first frame and why the junction between a reference clip and the generated speech matters (section 4).

Emotion labels in the natural data (SER top class AND speaker-normalised prosody agree): happy 5628, neutral 1688, disgust 506, angry 264, surprise 92, sad 69, fear 51 clips; 21,702 clips carry no confident emotion label. Fear, sad, surprise and angry are far too rare (tens to a few hundred clips) to learn a tag from natural data alone, which is why section 6 adds prosody-transformed samples.

4. The lost first word: what was measured and the root cause

User report: the male voice drops the beginning of the utterance ("ฉันชอบแมว" sounds like "..ชอบแมว"). Measured on v3 with the shipped voices, 30 onset sentences covering every initial sound class x 4 reference voices x 3 seeds (360 clips), judged by two independent Thai ASR models (Whisper large-v3-turbo and Pathumma-whisper-th-large-v3) and by the time from clip start to the first speech frame. A first word is 'heard' when the ASR text starts with the sentence's first word (PyThaiNLP newmm).

presentation	fw_turbo	fw_pathumma	t_first_ms
feminine	0.861	0.961	342.556
masculine	0.883	0.972	397.278

Finding 1: on the broad set the male voices are NOT worse than the female ones (first-word hit 0.97 masculine vs 0.96 feminine with Pathumma); the first word is lost in about 3 % of clips for every voice. Generated clips begin with 330-450 ms of silence on average, so the loss is not a clipped onset at sample 0 (level of the first 20 ms: below -75 dB).

Finding 2: the reported sentence reproduces, and it is specific to the app's default male reference (masculine_1). With the F5 default (50 ms of silence after the reference) masculine_1 loses "ฉัน" in 5 of 6 ASR judgements (3 seeds x 2 ASR); masculine_2 says it in all 6. That reference (masculine_1) is cut mid-sound (its last 50 ms are only 16 dB below

the clip's loud level; feminine_2 is similar), F5's own preprocessing trims reference silence to 50 ms, and the model continues the last sound into the new text.

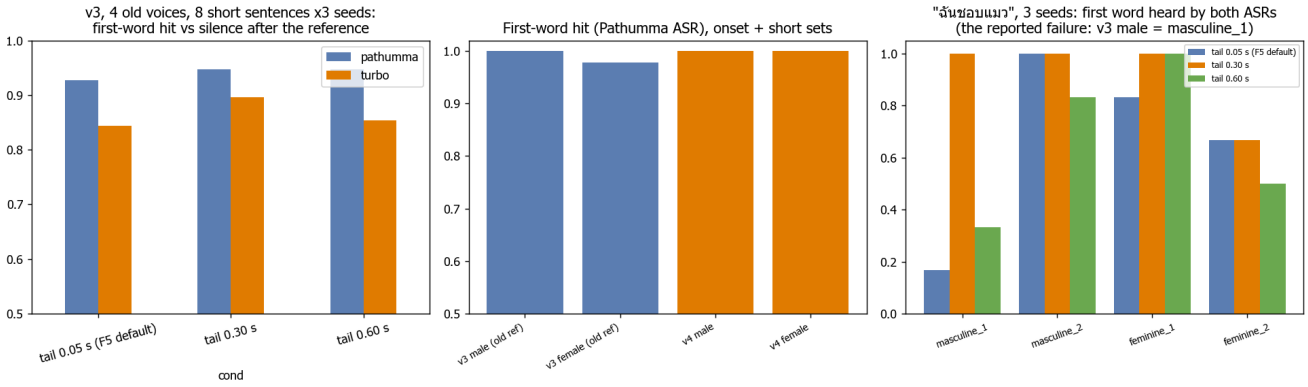


Figure 2. Left/centre: first-word hit vs silence after the reference and per voice; right: the reported sentence by voice and tail length (3 seeds, both ASR must hear the first word).

Ablation: appending silence after the reference (`ref_tail_s`) moved the overall first-word hit (8 short sentences x 4 voices x 3 seeds) from 0.927 to 0.948 (Pathumma) and 0.844 to 0.896 (turbo) with a 0.30 s tail; 0.60 s did not keep the gain (masculine_1's sentence fell back). With 3 seeds per cell this is suggestive, not conclusive. The padding is therefore a default (0.30 s), not the whole fix.

Finding 3 (largest effect): the choice of the reference. Eight youthful reference candidates (estimated age ≤ 25, clean ending, 4-9 s) were tried with v3 on the same short set: first-word hit (Pathumma) {'spk05ddc24a': 1.0, 'spk12fa307f': 1.0, 'spk1da492d7': 0.94, 'spk63668f9d': 1.0, 'spk6b483e05': 1.0, 'spk8f7bbabe': 1.0, 'spk99159c4a': 1.0, 'spkaf311c2d': 0.88} (per candidate speaker, anonymised) against 0.93-0.97 for the old voices; the four best became the new default voices and were held out of training.

5. Youthful voice (perceived age 18-23)

The data has no age labels. Perceived age was estimated with audeering's wav2vec2 age/gender regressor (CC-BY-NC-SA-4.0; trained on aGender, Common Voice, TIMIT, VoxCeleb 2 - Thai is not listed, so its validity on Thai is unverified: age correlates only weakly with speaker F0, r = -0.20, and the dataset's median estimate is 33 years). Results are called 'perceived/estimated age', never true age. 2,551 speakers have an estimate; 72 fall in 18-23 (43 masculine, 27 feminine).

Use: (a) youthful reference prompts were chosen among speakers estimated ≤ 25 whose clip ends cleanly (below -28 dB in the last 50 ms) and then by measured first-word hit; (b) training sampling weight 1 + exp(-0.5((age-21)/5)^2) (up to 2x for 16-26 years), as a sampling weight only - age is never a model input. The four default references are estimated at 23.6-24.3 years, i.e. just above the 18-23 target; this is reported as is.

Privacy and rights: the default voices are references cut from real dataset speakers (anonymised ids here; dataset licence CC-BY-3.0, weights derive from ThonburianTTS CC-BY-NC-SA). Consent of those speakers to voice cloning is unknown; treat the models as research / non-commercial.

6. Emotion control (7 FER2013 classes)

Mechanism (one, as planned): hybrid of a learned tag and prosody transforms. The natural data cannot teach the tags: emotion2vec_plus_large agreed with 3 of 14 old ElevenLabs emotion references, labelled 47 % of Thai clips neutral and 39 % 'happy' with almost no prosodic difference from neutral, and gives only a few dozen to a few hundred confident fear / sad / surprise / angry clips. Pseudo-labels therefore need a second signal: SER top class AND speaker-normalised prosody must agree. The shortfall is filled with prosody-transformed real clips (Praat PSOLA: F0 mean/range, rate, level; templates after Banse & Scherer 1996, strength 0.35-1 random per sample, label by construction); this is synthetic supervision, not human emotion data.

Evidence. Model tag alone (8 sentences x 7 emotions x 2 voices x 2 intensities): SER accuracy 0.152 / 0.152 (chance 0.143; angry recall 0.38, neutral 0.50, others ≤ 0.19), so the independent classifier does not recognise the emotions. Prosody (descriptive): level shifts have the right sign for happy/angry/sad/disgust (happy +0.8, angry +1.0, sad -1.7, disgust -0.9 dB at intensity 1) and rate follows the duration rule, but F0 shifts are 0.2-0.5 semitone against 1-3.5 in the templates. Cause: with a neutral prompt the DiT copies the prompt's pitch, and a few hundred updates of tagged prompt-format samples in a 20-minute run (the first 50-minute run was lost) were not enough to move it.

Mitigation shipped: tts.py applies the SAME PSOLA templates to the generated speech (config emotion_dsp, intensity-scaled, default 0.5). Measured on the neutral v4 clips: F0 shift at intensity 1.0 (semitones) - happy +1.6, sad -1.7, angry +1.1, fear +3.0, disgust -1.1, surprise +3.6 (templates: happy +1.5, sad -1.5, angry +1.0, fear +3.0, disgust -1.0, surprise +3.5), level/tempo as templated, CER 0.006-0.015. SER accuracy rises only to 0.20 / 0.25: the English-acted-speech classifier barely responds to ±1-3 semitone changes on Thai, so SER is a weak judge here. Naturalness of the PSOLA output is NOT measured: the blinded listening pack (data/run/gen_v4/listening_pack.html) is unrated.

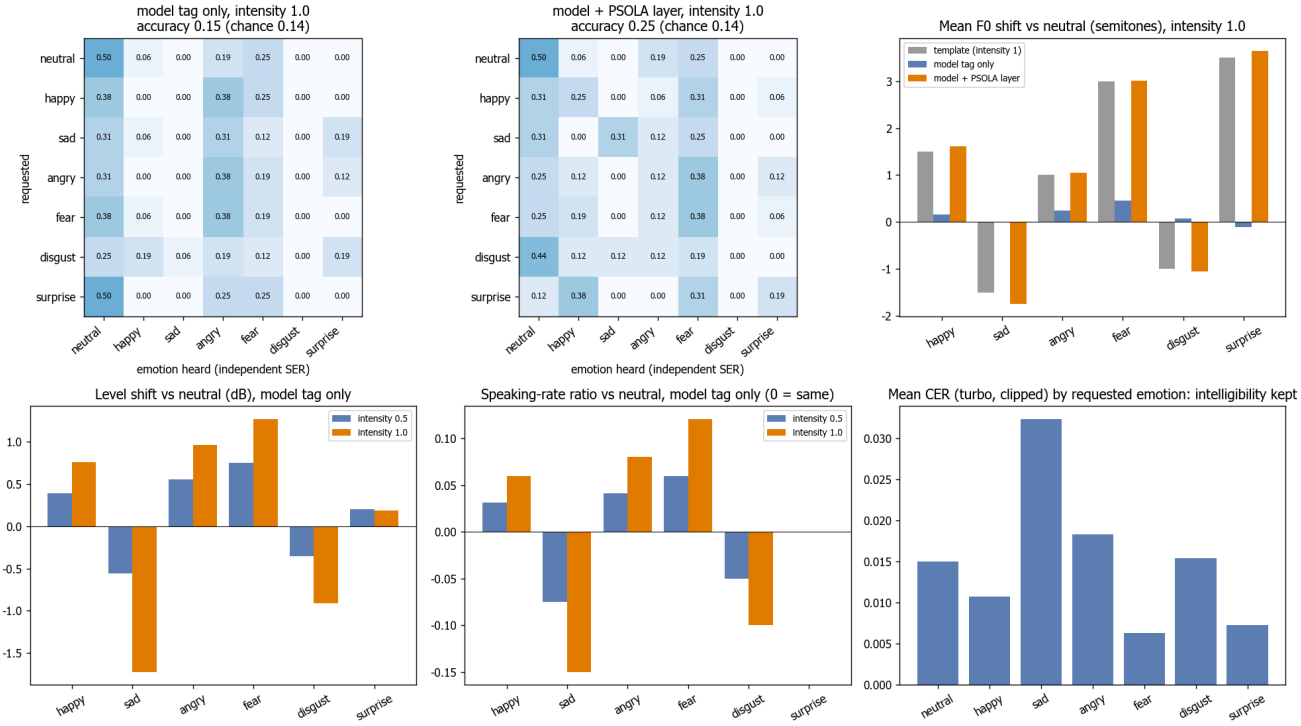


Figure 4. Emotion: confusion (model only vs model + PSOLA), F0 shift, level and tempo shifts, CER per emotion.

8. Latency

Measured first, then optimised with a quality gate each (median CER +0.01, speaker similarity -0.01, first-word hit not lower). 99 % of the time is the 32-step ODE (text prep 0.7 ms, vocoder 3.3 ms vs ODE 0.55 s on a 1 s clip, A100). Levers tested: EPSS step schedules (installed f5_tts 1.1.22 supports 5, 6, 7, 10, 12, 16 steps; verified in cfm.py/utils.py), shorter reference prompts, both on v3 (Figure 5, left) and on v4 (middle). Knee: 16 steps - first-word hit 1.00 (same as 32), mean CER 0.046 vs 0.037 at 32 steps (+0.009, inside the +0.01 gate), speaker similarity 0.865 vs 0.864; 12 steps fails (CER 0.107). 8 steps fails strongly on v3 (first-word hit 0.83, CER 0.44). Also: the two default references are 3-4 s (shorter prompt = fewer frames per step), both voices are pre-warmed at start-up, the reference tensors are cached, and a phrase cache answers repeats in ~10 ms.

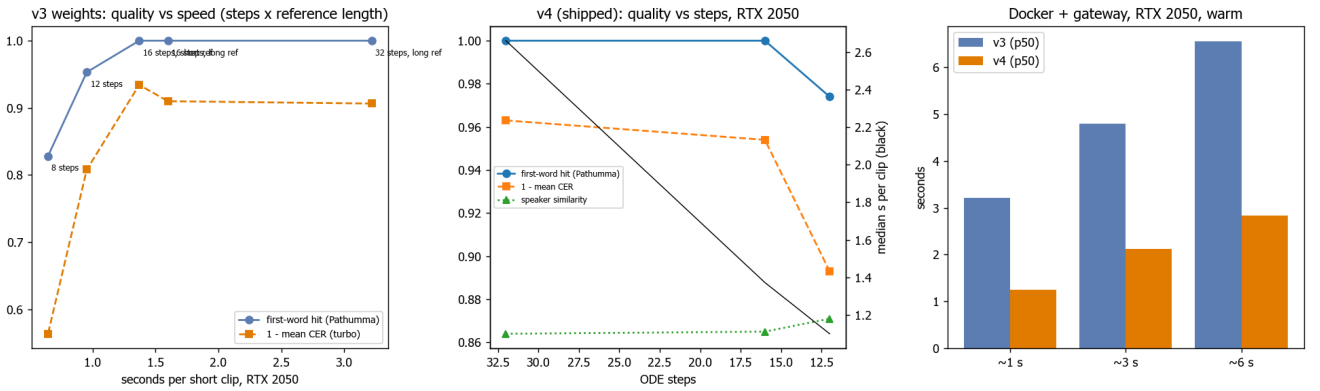


Figure 5. Left: v3 quality vs speed. Middle: v4 quality vs steps. Right: p50 latency v3 vs v4 in the real Docker stack.

Production (nginx gateway -> tts container, RTX 2050 shared with the desktop, warm, 10 requests per length): v3 3.21 / 4.80 / 6.55 s, v4 1.25 / 2.12 / 2.84 s (p95 within 0.15 s of p50); an emotional request adds ~0.1-0.2 s (PSOLA). A100

(run 1 job log, 16 vs 32 steps): v4 0.56 s vs v3 1.11 s per request regardless of length (GPU-launch bound). Cold start: v3 72 s, v4 68 s - not meaningfully reduced (model load dominates).

Application level (found while testing the stack): (1) sign-api's TTS_URL listed a non-existent 'tts-gpu' host first, so every request that attached audio waited ~5.3 s for that call to fail (6.2-10.7 s per request measured, 0.85 s after the fix); (2) the sign-api now returns the text immediately (TTS_DEFER=1, default) and the web fetches the audio with the existing speak() path - the API answered in ~0.75-0.9 s and the audio followed 1.2-1.8 s later; TTS_DEFER=0 restores v3 behaviour. Thresholds 0 / 0.5 / 0.8 behaved as before (all words / 'วันนี ...' / nothing to say), male and female voices and the speak button path were exercised through the real stack (data/stack_summary.json). Not done: torch.compile / CUDA graphs / TensorRT / distillation (documented as future work: costly and risky on a 4 GB GPU).

7b. Cloud run record

run	region	plan	result	billed	GPU cost	outcome
run 1	us-central1	50 min budget	3,014 updates, val 0.6331	97 min	$2.50	weights LOST (timeout during the output copy)
run 2	asia-southeast1	20 min budget	1,238 updates, val 0.6383	23.4 min	$0.60	shipped

7. Training

One planned recipe, run on a Vertex AI Spot A100 (the project's quota; no L4 quota; T4 cannot hold the batch and has no bf16). Starting point: Speech-model-v3-TTS (ThonburianTTS fine-tuned in v1). Changes against v1: (i) multi-grade pool and hours-balanced masculine/feminine sampling with a youth weight; (ii) PROMPT-FORMAT samples (50 % of items): [clip A of speaker S | clip B of S] with A as the prompt and the loss only on B, i.e. exactly the layout used at inference, with a random 0-120 ms prompt tail and B's leading silence normalised to 60-140 ms; (iii) loss weight 2 on the first 12 frames of the generated span; (iv) 12 % synthetic gap samples (unchanged from v1) so '…' still means a pause/hum; (v) emotion tags: seven unused vocabulary symbols placed before the text (no vocabulary change, checkpoint stays loadable), each tag embedding in its own parameter with a 20x learning rate, intensity = alpha of e_neutral + alpha(e_tag - e_neutral), 30 % of items carry a prosody-transformed emotion (section 6), 8 % a natural pseudo-label, 10 % the neutral tag, the rest no tag (the v3 path).

Run 1 (50 min budget, us-central1): 3,014 updates, best validation flow loss 0.6331; the job was cancelled by its own hard timeout while copying ~1,800 evaluation files, before the checkpoint reached the bucket (my script copied the checkpoint last, and the timeout counts from provisioning start): 97 min billed, ≈ $2.50, weights lost. Run 2 (20 min budget, asia-southeast1 only to avoid cross-region reads): 1,238 updates, 20.2 min, best validation loss 0.6383, 23.4 min billed, ≈ $0.60, model copied first. The shipped model is run 2. Validation loss (real + pair + gap items of held-out speakers, 30 % emotion-transformed) was nearly flat after ~1,200 updates in run 1 (0.6461 → 0.6331), which is why a 20-minute rerun was judged sufficient; loss does not measure emotion or onset, so the results below are what counts.

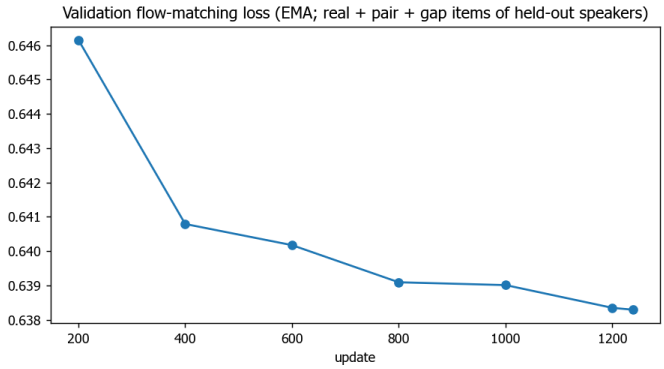


Figure 3. Validation flow-matching loss of the shipped run (EMA).

Cost (GPU, from the job records): run 1 $2.50 + run 2 $0.60 = $3.11. Not included in that figure (not visible in the job records): cross-region reads of the 3.9 GB upload for run 1 and storage, estimated at up to ~$0.5; the project total is therefore estimated at about $3.6 against the $4 ceiling. Both numbers are estimates, not a bill.

9. Results (v4 vs v3) and limits of the metrics

set	n (v4)	v3 CER med / mean	v4 CER med / mean	paired v4-v3 mean (95% CI)	v3 spk-sim	v4 spk-sim
sent	14	0.022 / 0.058	0.011 / 0.050	-0.008 (-0.059..+0.045)	0.938	0.944

set	n (v4)	v3 CER med / mean	v4 CER med / mean	paired v4-v3 mean (95% CI)	v3 spk-sim	v4 spk-sim
test	20	0.035 / 0.080	0.000 / 0.021	-0.059 (-0.111..-0.016)	0.939	0.952
word	50	0.062 / 0.354	0.000 / 0.286	-0.069 (-0.201..+0.069)	0.751	0.759
gap	6	0.000 / 0.109	0.019 / 0.199	+0.090 (-0.244..+0.500)	0.896	0.872
short	48	0.000 / 0.070	0.000 / 0.076	+0.006 (-0.045..+0.057)	0.840	0.793
onset	180	0.000 / 0.034	0.000 / 0.039	+0.005 (-0.011..+0.019)	0.879	0.885

CER = Whisper-large-v3-turbo, character edit distance, clipped at 1 (one clip looped to 2.05 in v1: median and clipped mean are reported); first-word hit = ASR text starts with the sentence's first word (PyThaiNLP newmm), Pathumma-whisper-th-large-v3 on seed 0 only (4 GB GPU), turbo on all seeds; speaker similarity = WavLM-base-plus-sv cosine to the voice's own reference (v3 and v4 use different references, so the similarity is comparable only as 'same level'); paired bootstrap (2,000 resamples) over the same sentence / seed / presentation, small n, descriptive. Emotion: evaluator SER = firdhokk/speech-emotion-recognition-with-openai-whisper-large-v3 (a different family from the labeller emotion2vec); it is trained on English acted speech and its validity on Thai is unknown. Single words under 0.9 s are unreliable for both models (v3 and v4 word-set CER 0.35 / 0.29 mean).

10. Limitations, licences, reproducibility, failures

Limitations: no human rating (MOS/CMOS) - the blinded pack is prepared and unrated; one training run (the first was lost), no training ablation (reference/tail/step effects were ablated at inference); small test sets; ASR/SER/age estimators validated on Thai only by indirect checks; perceived age is not true age; emotion training data is partly synthetic (PSOLA) and the model's own emotion control is weak; the PSOLA layer's naturalness is unmeasured; Pathumma first-word scores use seed 0 only; the v3 baseline rows from cloud run 1 used the new tail. Licences: weights derive from ThonburianTTS (CC-BY-NC-SA files in its repo) -> non-commercial / share-alike; dataset cc-by-3.0; the audeering age model is CC-BY-NC-SA-4.0; emotion2vec uses the FunASR model licence (not verified for commercial use). Default voices are references cut from real dataset speakers (consent unknown).

Reproducibility: prep.py select/fetch/label/build -> evaluate.py (refs are in models/Speech-model-v4-TTS/voices.json) -> vertex_tts.py upload/launch -> evaluate.py gen/score/affect/compare -> make_figures.py; env 490-TTS (+ funasr for the emotion labeller, parselmouth for PSOLA). Failures and cost: (1) run 1 lost at its timeout - the timeout counts from provisioning start and my script copied the checkpoint last; (2) first scoring attempts thrashed the 4 GB GPU (two ASR models resident) and were rewritten; (3) ElevenLabs Scribe could not be used as a third ASR (quota exhausted); (4) the first onset figure claimed a link between loud reference endings and lost first words that the broad data do not support - removed. GPU spend $3.11; estimated total ≈ $3.6 of the $4 ceiling.